

Find It



Campus Pickup Service System Design

Enterprise project

2021

National 3rd Prize in UXDA (International User Experience Innovation Competition)
Part of the system solution was adopted by Alibaba's courier service app team for experimental grayscale testing.



THE PROBLEM: CAMPUS PICKUP

People aged 19-30 make up 37% of online shoppers; students and young workers are key buyers. Campus pickup has distinct needs compared with other communities.

Many users
Many people and parcels
Sales events bring a surge

Busy pickup times
Similar pickup times
lead to congestion

Restricted access
Students must collect
their own parcels

KEY PROBLEMS TO SOLVE

Most couriers are moving toward smarter last-mile services. Cainiao serves 57% of campuses, yet satisfaction with pickup remains low. User research identified two main problems:

Slow to find
Parcels are hard to locate quickly
Poor shelf signs / crowded parcels / cluttered labels

Slow to check out
Checkout is complicated
Separate ID scans / queues at peak times

SOLUTION

RFID-powered
Efficiency at every step

01 A new campus pickup experience

Smart services at pickup stations
RFID packaging (green flexible printed electronics), parcel tracking, smart recycling bins, and creative reuse displays and exchanges

Online mini app
3D shelf and parcel previews, AR navigation, recycling rewards, social play

02 Parcel information in the cloud

Smarter shipping labels and protect personal information. Manage data by delivery stage (province, city, local area and last mile).

03 Recycling and social features

Use the campus community and points to reward recycling and reuse. Offer a place to exchange items, with fun features that add value.

DESIGN VALUE

A better pickup experience
Recycling awareness and social connection

CAMPUS PARCEL PICKUP TODAY

(Desk research, field visits and the first survey)

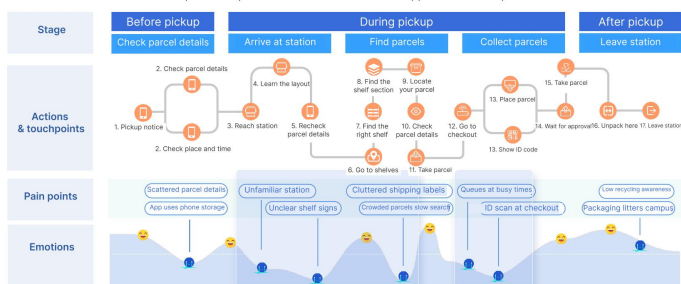
We studied campus pickup habits, problems and needs through field visits and a survey (24 universities; 270 valid responses).
Most couriers favor smart last-mile services. Cainiao serves 57% of campuses, but user satisfaction remains low.



USER RESEARCH

User Journey Map

Map each step: moments of frustration reveal opportunities to improve



USER RESEARCH

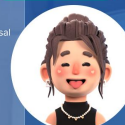
(Second survey and in-depth interviews)

Survey

A SWTH survey focused on pickup at Cainiao stations

Who: Gender, school, occupation, year, age, shopping frequency, box disposal
What: Satisfaction with pickup at Cainiao stations
Where: Pickup locations
When: Pickup times
Why: Reasons for low satisfaction
How: Suggestions for better campus pickup

64.6%



Liu Duoduo Female / 20 / Student
Frequent online shopper Collects parcels in batches
Picks up for classmates Tracks spending

Pain points

1. Many parcel types make searching difficult.
2. Each parcel needs a separate checkout.

User needs

1. Know exactly where each parcel is.
2. Collect several parcels faster.
3. Understand personal shopping habits.

In-depth Interviews

8 people with different profiles, all unhappy with Cainiao pickup
Explore pickup habits, feelings, good and bad experiences, and suggestions.

35.4%



Zhang Mangmang Male / 24 / Grad student
Shops for essentials Eco-conscious
Busy; dislikes queues

Pain points

1. Late classes often mean missing opening hours.
2. Long queues during busy periods.

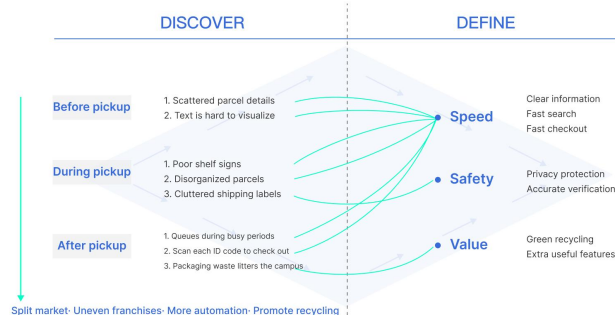
User needs

1. 24-hour pickup.
2. Faster pickup at peak times.
3. Safe, convenient pickup help.

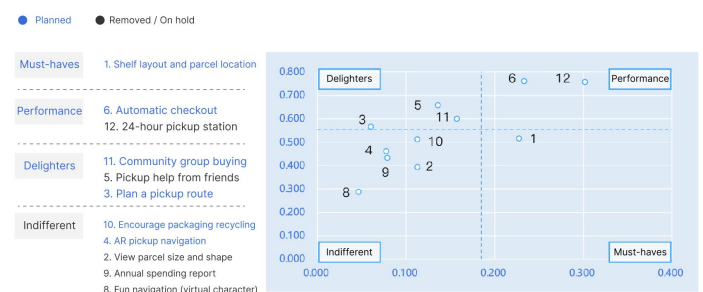
KEY USER NEEDS

DISCOVER

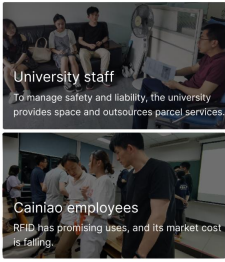
DEFINE



KANO NEEDS ANALYSIS

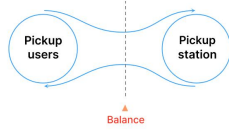


STAKEHOLDER INTERVIEWS



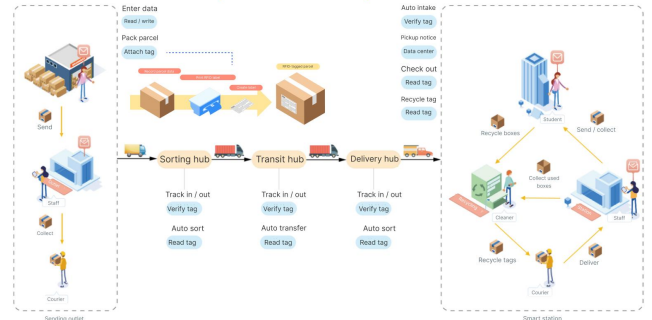
We interviewed university staff and Cainiao employees to explore solutions. For station managers, cost comes first. They often meet only basic pickup needs. As the industry becomes smarter, change is essential and user experience is receiving more attention.

When profit and user experience conflict, we need to balance station profitability with a better pickup experience.



SMART PARCEL PICKUP WITH RFID

A smarter system: faster delivery from start to finish



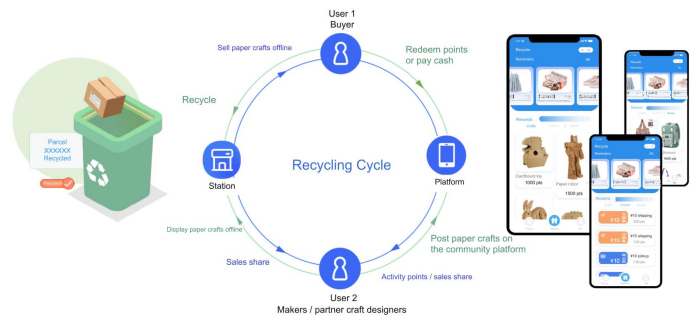
ENGAGEMENT STRATEGY

Use the campus community to encourage recycling and reuse



Recycling + Social Play

RECYCLING CYCLE

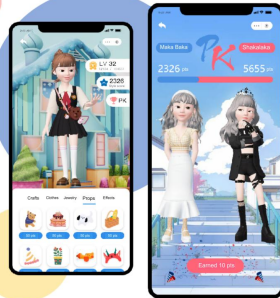


SOCIAL FEATURES

V-characters Interface

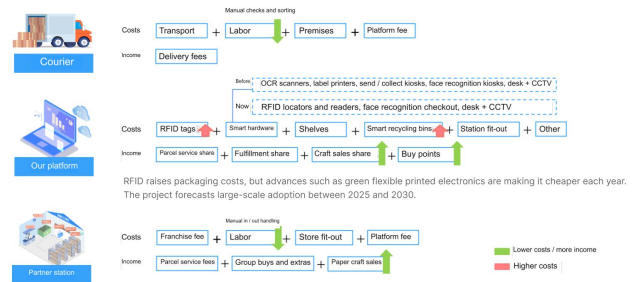
Shopping habits shape your profile and avatar. Use points to dress your avatar and raise your style score. Campus leaderboards let you compete with other students. Shake your phone to find nearby players for a style challenge. Winners have a chance to earn bonus rewards.

Social connection · Fun interaction · More spending · Repeat visits



COST AND REVENUE ANALYSIS

Lower labor costs, new ways to trade recycled goods, and revenue from points purchases



RFID raises packaging costs, but advances such as green flexible printed electronics are making it cheaper each year. The project forecasts large-scale adoption between 2025 and 2030.

Lower costs / more income
Higher costs



DELIVER

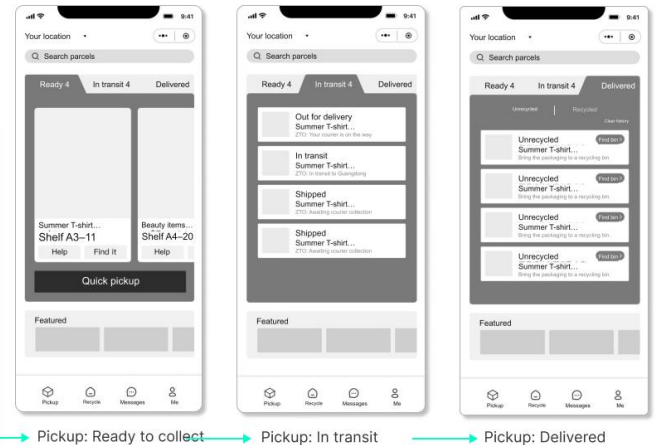
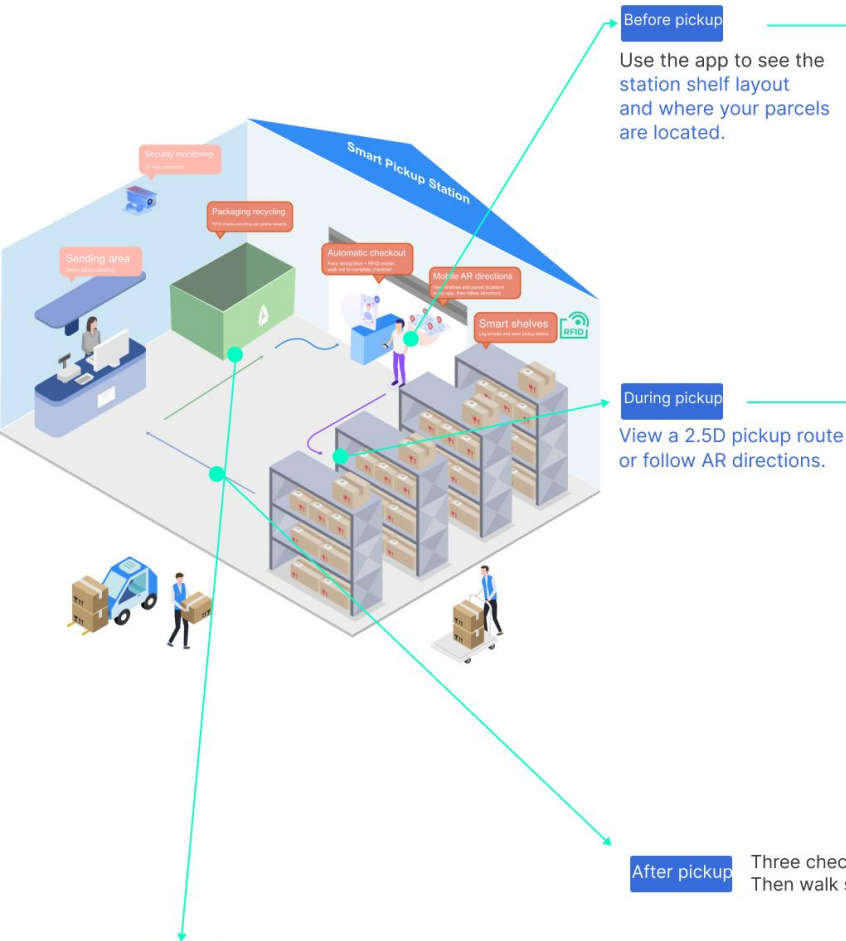
Smart Parcel Pickup with RFID

Smart station + Mini app = A better last-mile pickup experience

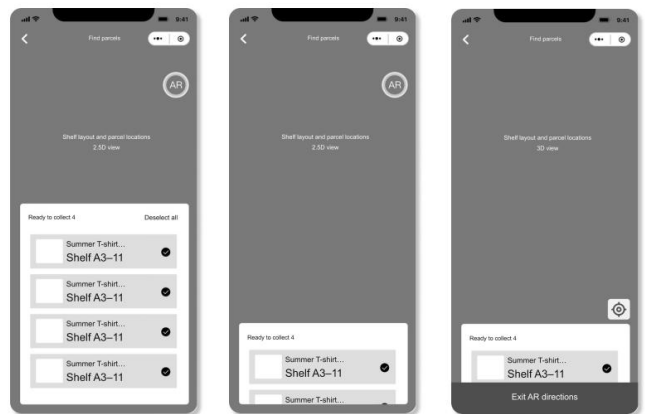
Pickup Station

User Actions

Key Interactions



2.5D route: List expanded → 2.5D route: List collapsed → AR directions



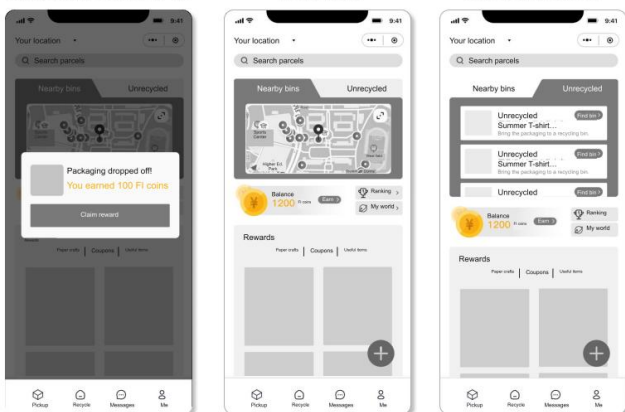
After pickup Three checks for accuracy: When you take a parcel, verify it is yours in the app. Then walk straight out of the station to finish checkout.

App: RFID + phone sensors verify ownership — self-check
Station: RFID reader + face recognition at the exit — theft prevention
Data: The shipping label includes a nickname and the last 4 phone digits

Recycling

Place RFID-tagged packaging in a recycling bin and earn rewards.

Recycling reward pop-up ← Nearby bins ← Unrecycled parcels



Wrong parcel alert ← Correct parcel toast

